

Practical-8 OUTPUT

1. Code Implementation:

Shows the C program written to implement the LRU page replacement algorithm, where memory frames are checked and the least recently used page is selected for replacement when a new page request occurs.



```
GNU nano 7.2 lru.c *
#include <stdio.h>

int main() {
    int frames = 3;
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int n = 12;

    int frame[3];
    int time[3];

    int pageFault = 0, pageHit = 0;
    int counter = 0;

    for(int i=0;i<frames;i++){
        frame[i] = -1;
    }

    for(int i=0;i<n;i++){
        int page = pages[i];
        int found = 0;

        for(int j=0;j<frames;j++){
            if(frame[j] == page){
                counter++;
            }
        }
    }
}
```

2. Compilation:

The GCC compiler converts the LRU C source code into an executable program so that the operating system can run the algorithm.



```
GNU nano 7.2 lru.c
        counter++;
        time[j] = counter;
        pageHit++;
        found = 1;
        break;
    }
}

if(found == 0){
    int pos = 0;

    if(frame[0] == -1){
        pos = 0;
    }
    else{
        int min = time[0];
        pos = 0;

        for(int j=1;j<frames;j++){
            if(time[j] < min){
                min = time[j];
                pos = j;
            }
        }
    }
}
```

3. Execution:

Runs the compiled program in the terminal to simulate the LRU page replacement process using the given page reference string and frame size.



```
GNU nano 7.2 lru.c
    pos = j;
    }
}
}

frame[pos] = page;
counter++;
time[pos] = counter;

pageFault++;
}

printf("Total Page Faults = %d\n",pageFault);
printf("Total Page Hits = %d\n",pageHit);

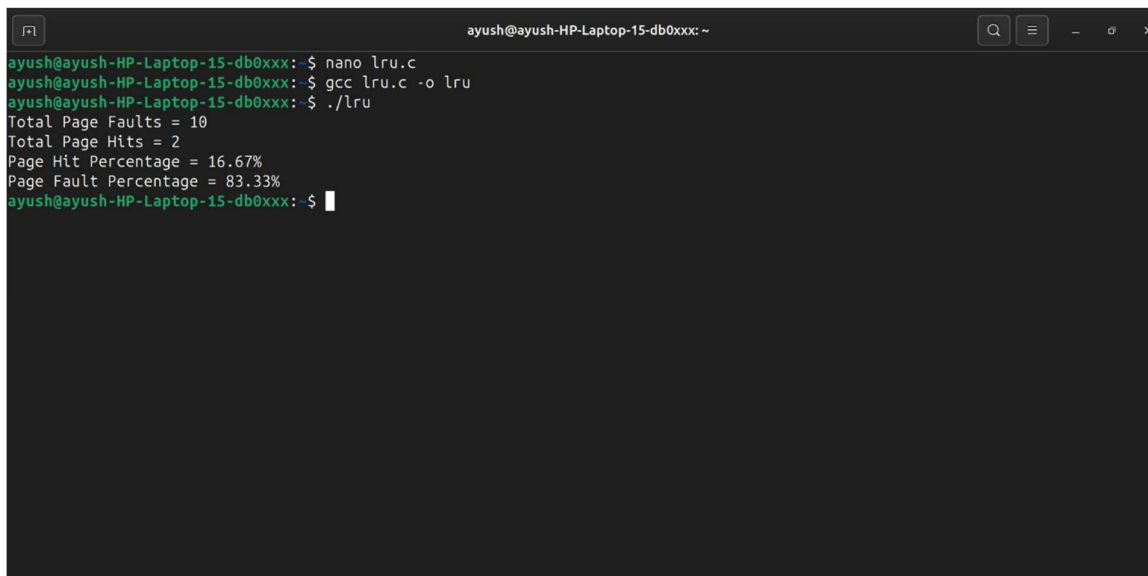
float hitPercent = (float)pageHit/n*100;
float faultPercent = (float)pageFault/n*100;

printf("Page Hit Percentage = %.2f%%\n",hitPercent);
printf("Page Fault Percentage = %.2f%%\n",faultPercent);

return 0;
}
```

4. Output:

Displays the calculated results of the LRU algorithm including total page hits, total page faults, and their respective percentages based on the given page reference sequence.



```
ayush@ayush-HP-Laptop-15-db0xxx: ~
ayush@ayush-HP-Laptop-15-db0xxx:~$ nano lru.c
ayush@ayush-HP-Laptop-15-db0xxx:~$ gcc lru.c -o lru
ayush@ayush-HP-Laptop-15-db0xxx:~$ ./lru
Total Page Faults = 10
Total Page Hits = 2
Page Hit Percentage = 16.67%
Page Fault Percentage = 83.33%
ayush@ayush-HP-Laptop-15-db0xxx:~$
```